



GIRRAWEEN HIGH SCHOOL

TASK 1 2022 (December 2021)

MATHEMATICS

EXTENSION 2

Complex Numbers

Time allowed – 90 Minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Each question attempted is to be marked clearly Question 6, Question 7 etc. Each question is to be returned on a separate page in your answer booklet.
- You may ask for spare answer booklets if you need them.
- For Multiple choice: Fill in the circle corresponding to the correct answer on the multiple choice answer sheet in your answer booklet.
- For Question 9(a) please draw your diagrams on the page of number planes provided in your answer booklet.
- The notation *cis* θ is acceptable in this examination.

Multiple Choice begins on the following page

Question 1

$$(-i)^{2022} =$$

- (A) i (B) $-i$ (C) 1 (D) -1

Question 2

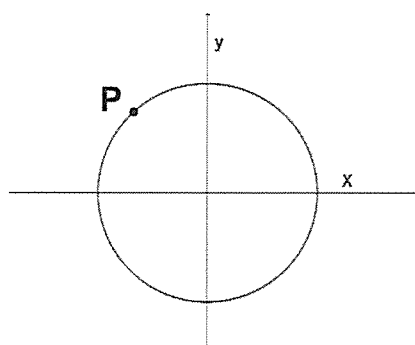
$$4e^{-\frac{\pi}{3}} =$$

- (A) $2\sqrt{3} - 2i$ (B) $2 - 2i\sqrt{3}$ (C) $-2\sqrt{3} + 2i$ (D) $-2 + 2i\sqrt{3}$

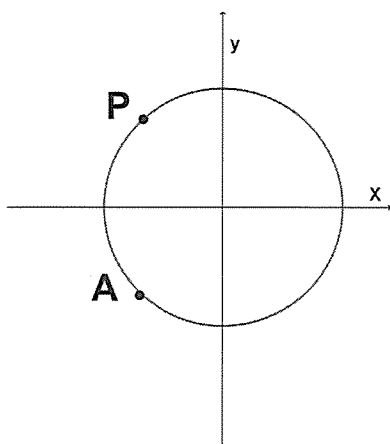
Question 3

If $\overrightarrow{OP} = z$, $\frac{\pi}{2} < \text{Arg } z < \pi$ and $|z| > 1$ (see diagram),

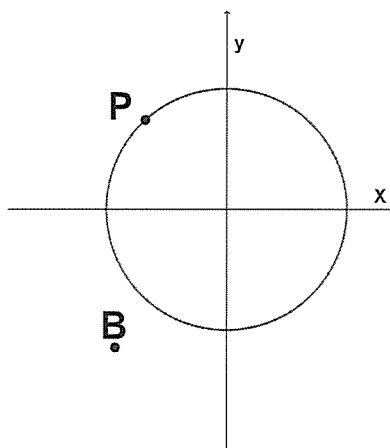
then $\sqrt{z}$ would be



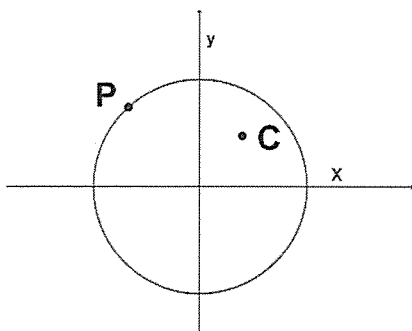
(A) $\overrightarrow{OA}$



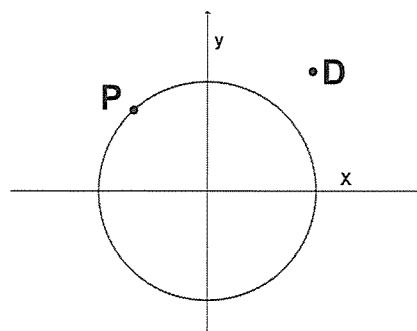
(B) $\overrightarrow{OB}$



(C) $\overrightarrow{OC}$



(D) $\overrightarrow{OD}$



Multiple Choice continues on the following page

Multiple Choice (continued)

Question 4

The solutions to $z^2 - 2z + 5 = 0$ are

- (A) $z = 1 \pm 2i$ (B) $z = -1 \pm 2i$ (C) $z = 1 \pm 4i$ (D) $z = 1 \pm 4i$

Question 5

$$z \times \frac{\bar{z}}{|z|^2} =$$

- (A) $\frac{z}{\bar{z}}$ (B) $\frac{1}{z}$ (C) z (D) 1

Question 6 (20 Marks – show all of your working in your answer booklet) **Marks**

(a) If $z = -\sqrt{3} - i$ and $w = 1 - i$

(i) Find $\frac{z}{w}$ in Cartesian form. **2**

(ii) Convert z and w to modulus/argument form. **4**

(iii) Find $\frac{z}{w}$ in modulus/argument form. **2**

(iv) Use your result to find the exact value of $\cos\left(-\frac{7\pi}{12}\right)$. **2**

(b) (i) By using $(x + iy)^2 = 15 - 8i$, find the square roots of $15 - 8i$ in Cartesian form. **3**

(ii) Hence solve $z^2 + (2 - i)z + (i - 3) = 0$. **3**

(c) Find the five fifth roots of $-16\sqrt{3} + 16i$. Leave your answers in modulus/argument form. **4**

Question 7 (7 Marks – show all of your working in your answer booklet)

(a) If the roots of the polynomial equation $4z^3 - 13z^2 + 40z + 39 = 0$ are α, β and λ , find $\alpha^2 + \beta^2 + \lambda^2$ WITHOUT finding α, β or λ . **2**

(b) Using your result from (a), state why $4z^3 - 13z^2 + 40z + 39 = 0$ has exactly one real root. **2**

(c) If one of the roots of $4z^3 - 13z^2 + 40z + 39 = 0$ is $z = 2 + 3i$, solve $4z^3 - 13z^2 + 40z + 39 = 0$ over the complex field. **3**

Examination continues on the following page

Question 8 (12 Marks – show all of your working in your answer booklet) **Marks**

(a) Sketch the following curves and regions on separate complex planes.

(i) $|z - 3 + i| = 2$ **2**

(ii) $\text{Arg}(z + 1) = -\frac{2\pi}{3}$ **2**

(iii) $|z| = 4$ and $-\frac{\pi}{4} < \text{Arg} z \leq \frac{\pi}{4}$ **3**

(b) The locus $\text{Arg}\left(\frac{z-3-i}{z+1-i}\right) = \frac{\pi}{4}$ represents part of a circle. Find the centre and radius of this circle and write the Cartesian equation for the locus. **5**

Question 9 (10 Marks – show all of your working in your answer booklet)

(a) If $\overrightarrow{OA} = z$ and $\overrightarrow{OB} = w$ with $0 \leq \text{Arg} z \leq \frac{\pi}{2}$,

$0 \leq \text{Arg} w \leq \frac{\pi}{2}$ and both $|z|$ and $|w| > 1$ (see diagram),

sketch on the Argand diagrams provided for 9(a) in your answer booklet

(Note that these vectors all start from O)

(i) C so that $\overrightarrow{OC} = z + w$ **1**

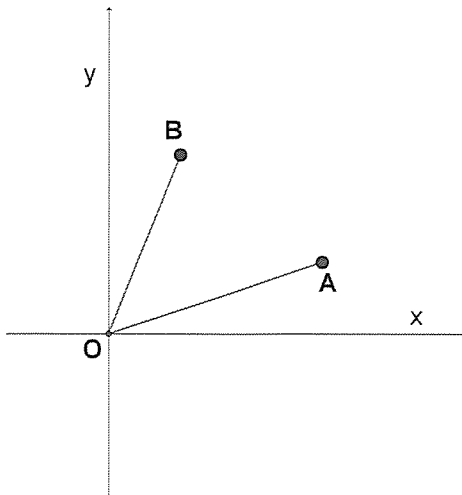
(ii) D so that $\overrightarrow{OD} = w - z$ **1**

(iii) E so that $\overrightarrow{OE} = z + 1$ **1**

(iv) F so that $\overrightarrow{OF} = z - 2i$ **1**

(v) G so that $\overrightarrow{OG} = -iz$ **1**

(vi) H so that $\overrightarrow{OH} = wcis\frac{\pi}{4}$ **1**

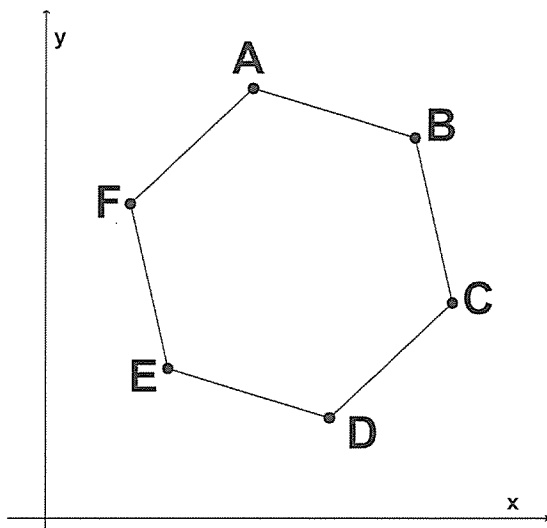


Question 9 continues on the following page

Question 9 (continued)

Marks

(b) $ABCDEF$ is a regular hexagon in Quadrant 1 of the complex plane
(see diagram)



If $\overrightarrow{OA} = z_1$, $\overrightarrow{OB} = z_2$, $\overrightarrow{OC} = z_3$, $\overrightarrow{OD} = z_4$, $\overrightarrow{OE} = z_5$ and $\overrightarrow{OF} = z_6$

- (i) Explain why $\text{Arg} \left(\frac{z_2 - z_3}{z_2 - z_5} \right) = \frac{\pi}{3}$ 2
- (ii) Show that $\left(\frac{z_2 - z_5}{z_3 - z_1} \right)$ is entirely imaginary. 2

Question 10 (13 Marks – show all of your working in your answer booklet)

- (a)(i) Solve $\cos 4\theta = 0, 0 \leq \theta \leq 2\pi$. 2
- (ii) Using DeMoivre's Theorem and the expansion of $(\cos\theta + i\sin\theta)^4$ 3
show that $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$.
- (iii) Hence find the exact value of $\cos \frac{\pi}{8}$. 2
(you may leave your answer as a fourth root).
- (b)(i) Using Euler's Theorem, show that $e^{ni\theta} + e^{-ni\theta} = 2 \cos n\theta$. 2
- (ii) Hence show that $\cos^4\theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$. 2
- (iii) Hence find the exact value of $\cos \frac{\pi}{8}$ 2
(you may leave your answer as a fourth root).

Examination continues on the following page

Question 11 (7 Marks – show all of your working in your answer booklet) **Marks**

(a) Solve $z^9 - 1 = 0$. **2**

(b) If w is the root of $z^9 - 1 = 0$ with the smallest positive argument, show that

(i) $w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = -1$ **1**

(you may assume that w^2, w^3 etc. are also roots of $z^9 - 1 = 0$).

(ii) Show that $(w^2 + w^7)(w^4 + w^5)(w + w^8) = -1$ **2**

(c) Hence show that $\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = \frac{1}{8}$. **2**

(You may assume that if $z = \text{cis}\theta$, $z^n + \frac{1}{z^n} = 2\cos n\theta$)

END OF EXAMINATION!!!



GIRRAWEE HIGH SCHOOL

MATHEMATICS EXTENSION 2

2022 HSC Task 1 (December 2021)

Student No: _____

Solution [Final]

This Booklet contains the answer sheet for Part A and Writing Booklet for Part B.

Part A ANSWER SHEET

Select the alternative A, B, C or D that best answers the question.

1.	A	☐	B	☐	C	☐	D	☒
2.	A	☐	B	☒	C	☐	D	☐
3.	A	☐	B	☐	C	☒	D	☐
4.	A	☒	B	☐	C	☐	D	☐
5.	A	☐	B	☐	C	☐	D	☒

Instructions

- If you need more paper for Part B, please ask your supervisor.
- Write your name on every booklet you use.
- Write on both sides of each sheet of paper.

Total number of booklets used _____.

Solution: Complex Numbers Test: Dec. 2021 p.1

Multiple Choice:

(1) $(-i)^{2022}$

$$= (-i)^{2020} \times (-i)^2$$

$$= 1 \times -1$$

$$= -1. \quad \textcircled{D}$$

(2) $4e^{-\frac{\pi}{3}}$

$$= 4 \left[\cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right) \right]$$

$$= 4 \left[\frac{1}{2} - i \frac{\sqrt{3}}{2} \right]$$

$$= 2 - 2i\sqrt{3}. \quad \textcircled{B}$$

(3) $|\sqrt{z}| < |z|$ if $|z| > 1$

so $\sqrt{z}$ is INSIDE circle.

If $\frac{\pi}{2} < \text{Arg } z < \pi$ then

$$0 < \text{Arg}(\sqrt{z}) < \frac{\pi}{2}$$

as $\text{Arg}(\sqrt{z}) = \frac{1}{2} \text{Arg } z. \quad \textcircled{C}$

(4) $z^2 - 2z + 5 = 0$

$$(z-1)^2 + 4 = 0$$

$$[z-1-2i][z-1+2i] = 0$$

$$z = 1 \pm 2i. \quad \textcircled{A}$$

[Could also be done using the quadratic formula].

(5) $z \times \frac{\bar{z}}{|z|^2}$

$$= \frac{z\bar{z}}{|z|^2}$$

$$= \frac{|z|^2}{|z|^2}$$

$$= 1 \quad \textcircled{D}$$

Q. (6)(a)(i) $\frac{z}{w}$

$$= \frac{\sqrt{3}-i}{1-i} \times \frac{(1+i)}{(1+i)}$$

$$= \frac{-\sqrt{3}-i\sqrt{3}-i+1}{2}$$

$$= \frac{(1-\sqrt{3})-i(\sqrt{3}+1)}{2}$$

$$= \left(\frac{1-\sqrt{3}}{2} \right) + i \left(\frac{-\sqrt{3}-1}{2} \right)$$

(ii) $z = 2 \text{cis}\left(-\frac{5\pi}{6}\right)$ 4

$$w = \sqrt{2} \text{cis}\left(-\frac{\pi}{4}\right)$$

(iii) $\frac{z}{w} = \sqrt{2} \text{cis}\left(-\frac{7\pi}{12}\right)$ 2

(iv) Equating real parts of (i) & (iii)

$$\frac{1-\sqrt{3}}{2} = \sqrt{2} \cos\left(-\frac{7\pi}{12}\right) \quad \underline{\underline{2}}$$

$$\therefore \cos\left(-\frac{7\pi}{12}\right) = \frac{1-\sqrt{3}}{2\sqrt{2}}$$

(b)(i) $(x+iy)^2 = 15 - 8i, x, y \text{ REAL}$

$$(x^2 - y^2) + 2ixy = 15 - 8i$$

Equating reals, Equating imaginaries

$$x^2 - y^2 = 15 \quad (1) \quad 2xy = -8$$

$$y = -\frac{4}{x} \quad (2)$$

Sub. (2) in (1):

$$x^2 - \left(-\frac{4}{x}\right)^2 = 15$$

$$x^2 - \frac{16}{x^2} = 15$$

$$x^4 - 15x^2 - 16 = 0$$

$$(x^2 - 16)(x^2 + 1) = 0$$

As x is real, $x^2 - 16 = 0$
 $x = \pm 4.$

As $y = -\frac{4}{x}$ 3

$$y = -\frac{4}{4} \text{ or } -\frac{4}{-4}$$

$$= -1 \text{ or } 1.$$

So $\sqrt{15-8i}$

$$= \pm(4-i).$$

Complex Numbers: p.2

$$Q.(6)(b)(i) \quad z^2 + (2-i)z + (i-3) = 0$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2+i \pm \sqrt{15-8i}}{2 \times 1}$$

$$= \frac{-2+i \pm (4-i)}{2} \text{ [from (b)(i)]}$$

Note: $b^2 - 4ac$

$$= (2-i)^2 - 4 \times 1 \times (i-3)$$

$$= 4 - 4i - 1 - 4i + 12$$

$$= 15 - 8i$$

3

$$z = \frac{-2+i+4-i}{2}$$

$$\text{or } z = \frac{-2+i-4+i}{2}$$

$$z = 1 \quad \text{or } z = -3+i$$

$$(c) \quad -16\sqrt{3} + 16i$$

$$= 32 \operatorname{cis} \frac{5\pi}{6}$$

$\therefore$ The five 5th roots of $-16\sqrt{3} + 16i$

$$= 2 \operatorname{cis} \left[\frac{\pi}{6} + \frac{2k\pi}{5}, k=0,1,2,3,4 \right]$$

4

$$\text{or } = 2 \operatorname{cis} \frac{\pi}{6}, 2 \operatorname{cis} \frac{17\pi}{30}, 2 \operatorname{cis} \frac{29\pi}{30}, 2 \operatorname{cis} \frac{41\pi}{30}, 2 \operatorname{cis} \frac{53\pi}{30}$$

$$\text{or } = 2 \operatorname{cis} \frac{\pi}{6}, 2 \operatorname{cis} \frac{17\pi}{30}, 2 \operatorname{cis} \frac{29\pi}{30}, 2 \operatorname{cis} \frac{-19\pi}{30}, 2 \operatorname{cis} \frac{-7\pi}{30}$$

Complex Numbers Solutions [continued] p.3

$$\begin{aligned} \text{Q. (7) (a)} \quad & \alpha^2 + \beta^2 + \lambda^2 \\ &= (\alpha + \beta + \lambda)^2 - 2(\alpha\beta + \alpha\lambda + \beta\lambda) \\ &= \left(\frac{13}{4}\right)^2 - 2 \times \frac{40}{4} \\ &= \underline{\underline{-\frac{151}{16}}} \end{aligned}$$

(b) As $\alpha^2 + \beta^2 + \lambda^2 < 0$, $4z^3 - 13z^2 + 40z + 39 = 0$ must have at least 1 non-real root as a real number squared ≥ 0 . 2

As $4z^3 - 13z^2 + 40z + 39 = 0$ has all real co-efficients, the CONJUGATE of any non-real root must ALSO be a solution.

Hence $4z^3 - 13z^2 + 40z + 39 = 0$ has 1 real & 2 non-real roots.

(c) One root of $4z^3 - 13z^2 + 40z + 39 = 0$ is $z = 2 + 3i$.
 $\therefore$ Another root is $\bar{z} = 2 - 3i$.

$$\text{So by } \alpha + \beta + \lambda = -\frac{b}{a}.$$

$$\begin{aligned} (2+3i) + (2-3i) + \lambda &= \frac{13}{4} \\ \lambda &= \underline{\underline{-\frac{3}{4}}} \end{aligned}$$

Solutions to $4z^3 - 13z^2 + 40z + 39 = 0$ are
 $z = 2 \pm 3i, -\frac{3}{4}$

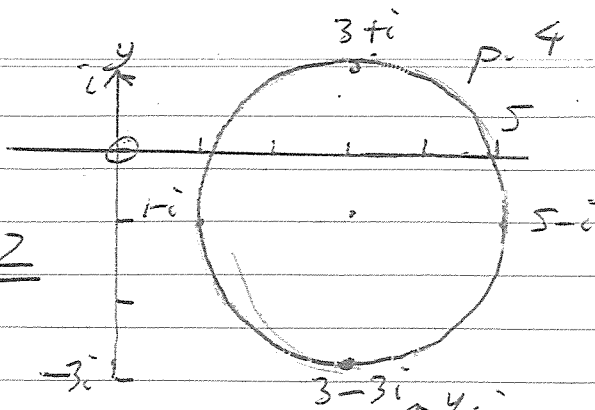
Q. (8) (a)

(i) $|z - 3 + i| = 2$.

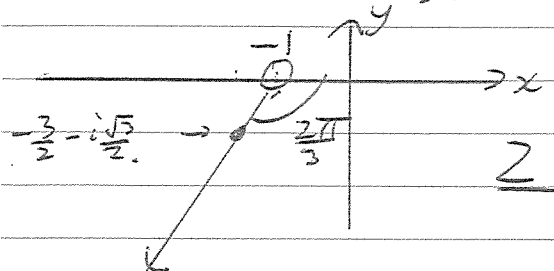
Circle centre $3 - i$, $r = 2$.

Equation:

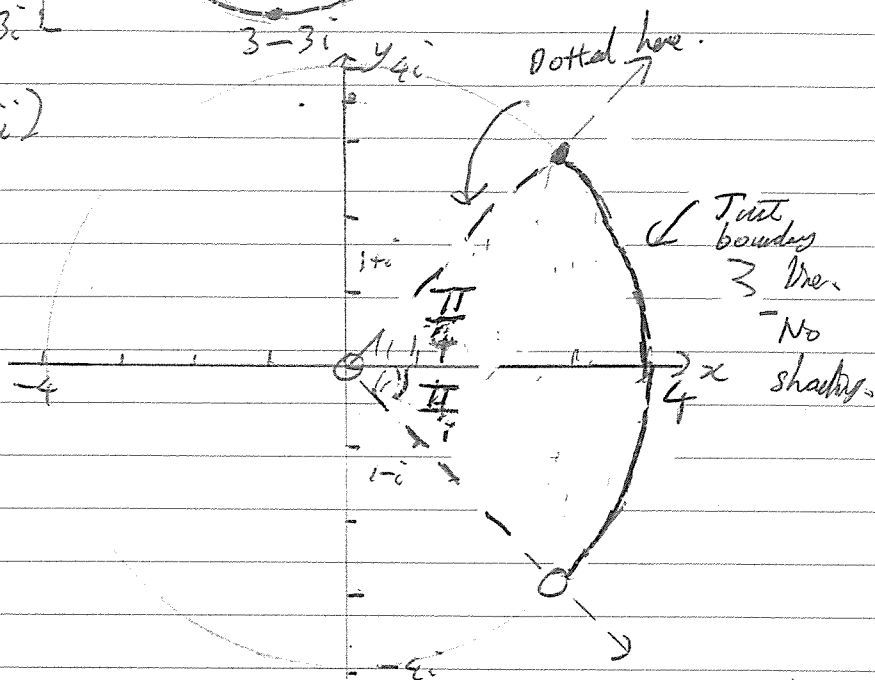
$(x - 3)^2 + (y + 1)^2 = 4$. 2



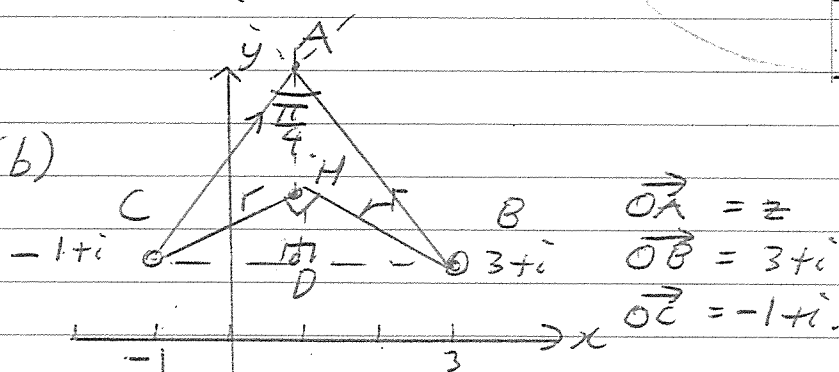
(ii) $\text{Arg}(z + 1) = -\frac{2\pi}{3}$.



(iii)



Q. (8) (b)



$\vec{OA} = z$

$\vec{OB} = 3 + i$

$\vec{OC} = -1 + i$

$\vec{BA} = z - 3 - i$ $\frac{\pi}{2} \leq \text{Arg } \vec{BA} < \pi$

$\vec{CA} = z + 1 - i$ $0 \leq \text{Arg } \vec{CA} < \frac{\pi}{2}$

$\therefore \text{Arg} \left(\frac{z - 3 - i}{z + 1 - i} \right) = \angle CAB = \frac{\pi}{4}$

Let circle centre = H.

x co-ordinate of H = 1.

$\angle CHB = \frac{\pi}{2}$ ($\angle$ at centre = $2 \times \angle$ at circumference subtended by arc CB).

$CH = BH = r$ [radius of circle]

By Pythagoras on $\triangle CHB$, $r^2 + r^2 = (CB)^2 = 4$.
 $r = 2\sqrt{2}$.

Let D be midpoint of

$HCB = 1 + i$

Using $\triangle CHO$:

$2\sqrt{2}^2 = (2\sqrt{2})^2 - 2^2 = 4$.

$CH^2 = 4$. 5

$\therefore H = 1 + 3i$

Locus is

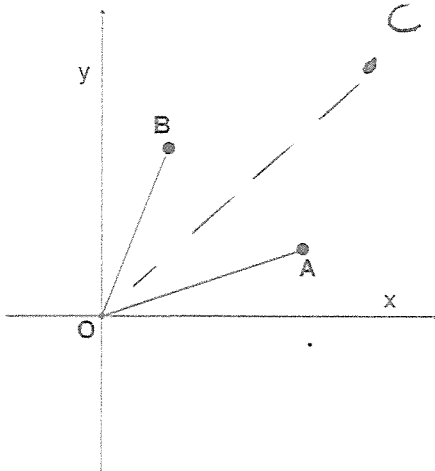
$(x - 1)^2 + (y - 3)^2 = 8$,

$y > 1$.

P.5

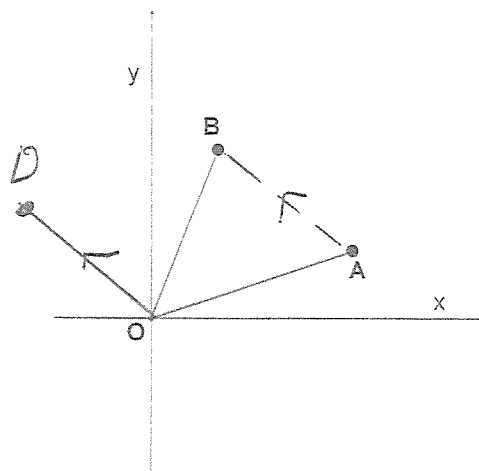
Answer sheet for Question 9(a)

(i)



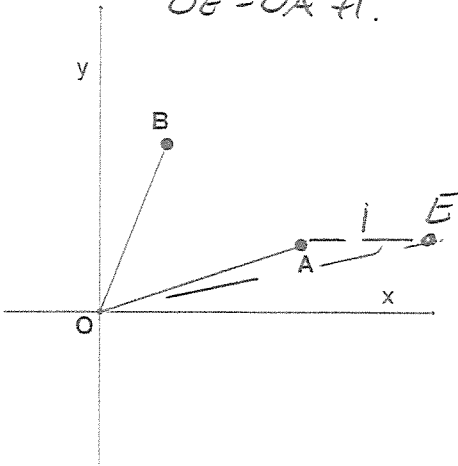
(ii)

$$\vec{OD} = \vec{AB} = w - z.$$

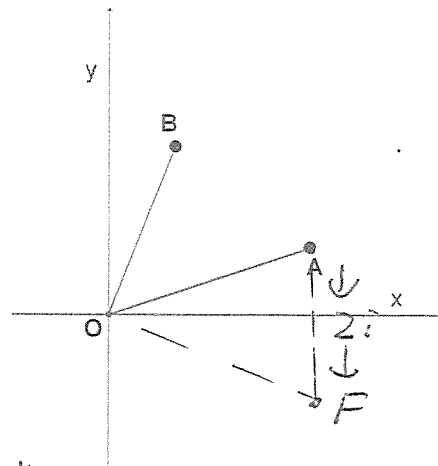


(iii)

$$\vec{OE} = \vec{OA} + \mathbf{i}.$$

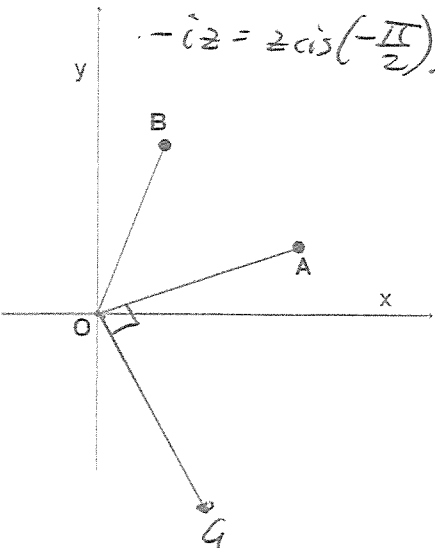


(iv)

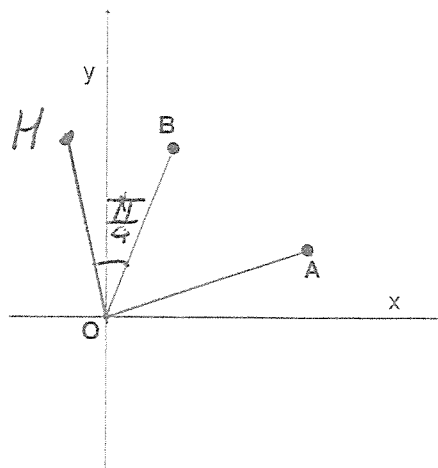


(v)

$$-iz = z \cos\left(-\frac{\pi}{2}\right).$$

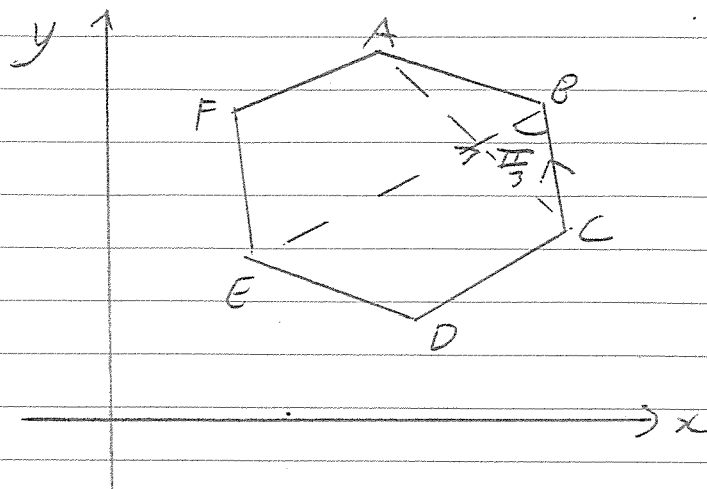


(vi)



Q. (9) (b)

(p. 6)



$$(i) \operatorname{Arg} \left(\frac{z_2 - z_3}{z_2 - z_5} \right) = \angle EBC \text{ as } z_2 - z_3 = \vec{CB} \\ \& \ z_2 - z_5 = \vec{EB}.$$

$$\angle ABC = \frac{2\pi}{3} \text{ (interior } \angle \text{ of regular hexagon).} \quad \underline{2} \\ \text{As EB bisects } \angle ABC, \angle EBC = \frac{\pi}{3}$$

$$(ii) z_3 - z_1 = \vec{AC}$$

Letting G be the point of intersection of AC & EB:

Using $\triangle BCO$

$$\angle BCA = \frac{\pi}{6} \text{ [} \angle \text{ opposite = sides in isosceles } \triangle ABC \text{]}$$

Using $\triangle BCG$

$$\angle BGC = \frac{\pi}{2} \text{ [} \angle \text{ sum of } \triangle BGC \text{]} \quad \underline{2}$$

$$\therefore \text{ As } \operatorname{Arg} \left(\frac{z_2 - z_5}{z_3 - z_1} \right) = \angle BGC.$$

$$\operatorname{Arg} \left(\frac{z_2 - z_5}{z_3 - z_1} \right) = \frac{\pi}{2}.$$

$$\frac{z_2 - z_5}{z_3 - z_1} \text{ is entirely imaginary.}$$

p. 7

Q. (10) (a) (i) $\cos 4\theta = 0$

$$4\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}, \frac{11\pi}{2}, \frac{13\pi}{2}, \frac{15\pi}{2}$$

$$\theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8} \quad \underline{2}$$

(ii) By de Moivre

$$(\cos \theta + i \sin \theta)^4$$

$$= \cos 4\theta + i \sin 4\theta$$

$$\cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta = \cos 4\theta + i \sin 4\theta \quad \text{--- (1)}$$

Equating real parts,

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$= \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$$

$$= 8 \cos^4 \theta - 8 \cos^2 \theta + 1, \text{ as required.} \quad \underline{3}$$

(iii) Hence $\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$ etc. are solutions to $\cos 4\theta = 0$
i.e. $8 \cos^4 \theta - 8 \cos^2 \theta + 1 = 0$

$$\cos^2 \theta = \frac{8 \pm \sqrt{(-8)^2 - 4 \times 8 \times 1}}{2 \times 8}$$

$$\cos^2 \theta = \frac{2 \pm \sqrt{2}}{4}$$

$$\cos^2 \theta = \pm \sqrt{\frac{2 \pm \sqrt{2}}{4}} \quad \underline{2}$$

As $\frac{\pi}{8}$ is in Q_1 , $\cos \frac{\pi}{8}$ will be positive &

will be $\sqrt{\frac{2+\sqrt{2}}{4}}$ or $\sqrt{\frac{2-\sqrt{2}}{4}}$.

$$\text{As } \cos \frac{\pi}{8} > \cos \frac{3\pi}{8}, \cos \frac{\pi}{8} = \sqrt{\frac{2+\sqrt{2}}{4}}$$

Q. 10(b)(i) By Euler's theorem,

$$e^{ni\theta} + e^{-ni\theta} = (\cos n\theta + i\sin n\theta) + (\cos(-n\theta) + i\sin(-n\theta)) \quad \underline{2}$$

$$= \cos n\theta + i\sin n\theta + \cos n\theta - i\sin n\theta \quad [\text{as } \cos \text{ even, } \sin \text{ odd}]$$

$$= 2\cos n\theta \quad \text{as required.}$$

(ii) Hence $[2\cos\theta]^4 = (e^{i\theta} + e^{-i\theta})^4$

$$16\cos^4\theta = e^{4i\theta} + 4e^{2i\theta} + 6 + 4e^{-2i\theta} + e^{-4i\theta} \quad \underline{2}$$

$$= (e^{4i\theta} + e^{-4i\theta}) + 4(e^{2i\theta} + e^{-2i\theta}) + 6$$

$$16\cos^4\theta = 2\cos 4\theta + 8\cos 2\theta + 6$$

$$\cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8} \quad \text{as required.}$$

(iii) Hence $\cos^4\frac{\pi}{8} = \frac{1}{8}\cos\frac{\pi}{2} + \frac{1}{2}\cos\frac{\pi}{4} + \frac{3}{8}$

$$\cos^4\frac{\pi}{8} = \frac{1}{2\sqrt{2}} + \frac{3}{8} \quad \underline{2}$$

$$\cos^4\frac{\pi}{8} = \frac{2\sqrt{2} + 3}{8}$$

$$\cos\frac{\pi}{8} = \sqrt[4]{\frac{2\sqrt{2} + 3}{8}}$$

as cos positive in Q_1 , just positive $\sqrt[4]{}$.

[Note: Can also do $\cos^4\frac{\pi}{8} = \frac{4\sqrt{2} + 6}{16}$

$$\cos\frac{\pi}{8} = \sqrt[4]{\frac{4\sqrt{2} + 6}{16}}$$

p. 9

Q.11)(a) $z^9 - 1 = 0$

$z = \text{cis } \frac{2k\pi}{9}, k = 0, 1, 2, 3, 4, 5, 6, 7, 8.$

(b)(i) $z^9 - 1 = 0$

By $\alpha + \beta + \gamma + \lambda = -\frac{b}{a}$

$1 + w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = 0$

$w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = -1.$

Note: Can also use

$z^9 - 1 = 0$

$(z-1)(1+z+z^2+z^3+z^4+z^5+z^6+z^7+z^8) = 0$

sub in w
 $(w-1)(1+w+w^2+w^3+w^4+w^5+w^6+w^7+w^8) = 0$

As $w \neq 1$, $1 + w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = 0.$

(ii) $(w^2 + w^7)(w^4 + w^5)(w + w^8)$

$= (w^6 + w^7 + w^{11} + w^{12})(w + w^8)$

$= w^7 + w^{14} + w^8 + w^{15} + w^{12} + w^{19} + w^{13} + w^{20}$

$= w^7 + w^5 + w^8 + w^6 + w^3 + w + w^4 + w^2$

$= -1$ (using $f(i)$).

(c) As $w + w^8 = \text{cis } \frac{2\pi}{9} + \text{cis } \frac{16\pi}{9}$

$= 2 \cos \frac{2\pi}{9}$

Similarly $w^2 + w^7 = 2 \cos \frac{4\pi}{9}$ & $w^4 + w^5 = 2 \cos \frac{8\pi}{9}$

$= -2 \cos \frac{\pi}{9}$

By $(w^2 + w^7)(w^4 + w^5)(w + w^8)$

$= -1.$

$2 \cos \frac{4\pi}{9} \times 2 \cos \frac{\pi}{9} \times 2 \cos \frac{2\pi}{9}$

$= -1$

$-8 \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}$

$= -1 \therefore -8$

$\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}$

$= \frac{1}{8}$

END OF SOLUTION!!!